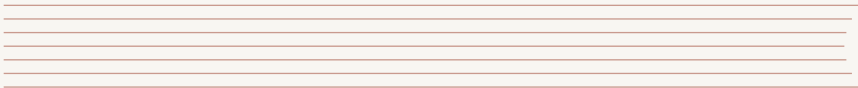


DATA WAREHOUSE

Air Quality, Economy and Health in Europe (2010–2020)

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midcentury
modern



OUTLINE

- Introduction and Dataset
- Conceptual Design (DFM)
- Starflake Logical Schema
- Reconciled Layer: 3-Tier ELT
- OLAP Analysis and Insights
- Conclusions

■ The starting idea

Is there an observable relationship between **air pollution**, **economic wealth**, and **mortality** across European countries? Answering this requires a schema able to connect different domains on the same geographic and time basis — this is why we chose a Data Warehouse.

Environment (Air Quality), Economy (GDP), and Health (Mortality) are usually analyzed separately: combining them into a single multidimensional model enables comparisons that would otherwise be impossible — for example, understanding whether a richer country also breathes cleaner air and lives longer. Ten-year coverage across Europe (2010–2020).

DOMAINS AND DATA SOURCES

Area	Source	Files	Format
Air Quality	WHO	1 (city, PM2.5/PM10/NO2)	Excel
Mortality (HFAMDB)	WHO	4 (<i>All causes</i> + 3 specific causes)	CSV
Economy (GDP)	Eurostat	1	TSV
Population	Eurostat	1	TSV

Even data from the same source come in different formats (GDP and Population are 2 separate Eurostat files, not one): structures, country codes, and granularity do not match — a cleaning and integration step is needed before loading them into the same Data Warehouse.

A CLOSER LOOK AT THE SOURCE FIELDS

WHO — Air Quality (Excel) (2000–2021)

IS03, WHO Country Name, City or Locality, Measurement Year, PM2.5, PM10, NO2, % time coverage — one row per (city, year).

Eurostat — GDP, estat_nama_10_pc.tsv (1975–2025)

freq, unit, na_item, geo, then one year per column: *wide* format.

Eurostat — Population, estat_demo_pjan.tsv (1960–2025)

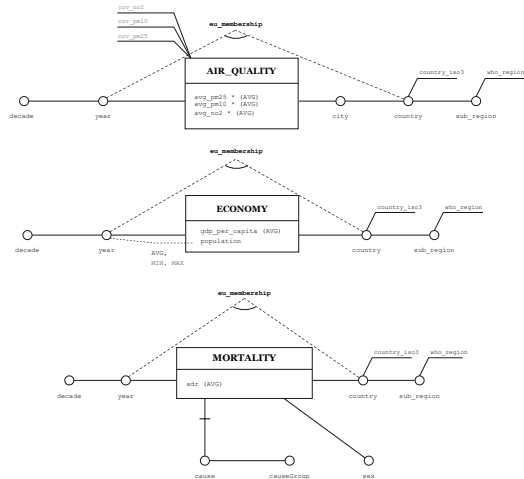
freq, unit, age, sex, geo, then one year per column: a separate file from GDP, with different columns — they are only joined at the reconciled stage.

WHO — HFAMDB Mortality (CSV, 4 files) (1968–2022)

COUNTRY, COUNTRY_GRP, AGE_GRP_LIST, SEX, SUBNATIONAL_MDB, YEAR, VALUE: one row per (country, sex, year), one file per cause of death.

THE CONCEPTUAL MODEL (DFM)

- **Air Quality:** granularity City / Year. Measures include PM2.5, PM10, and NO2 levels, along with their respective temporal coverage.
- **Economy:** granularity Country / Year. Measures include GDP per capita and population.
- **Mortality:** granularity Country / Year / Gender / Cause. The measure is the Standardized Death Rate (SDR), it is aggregated with AVG, never with SUM. CAUSE is an optional dimension: populated only by the specific diseases.



Why eu_membership is Cross-Dimensional

In our Conceptual Model (DFM), eu_membership is designed as a **cross-dimensional attribute** rather than a static dimensional property. It depends on the **combination** of Country and Year, since EU membership is not static but changes over time. This design choice allows us to accurately capture historical transitions within our 2010–2020 observation window.

Country	Year	eu_membership
HRV (Croatia)	2012 → 2013	Non-EU → EU
GBR (UK)	2019 → 2020	EU → Non-EU

GRANULARITY AND BUS MATRIX

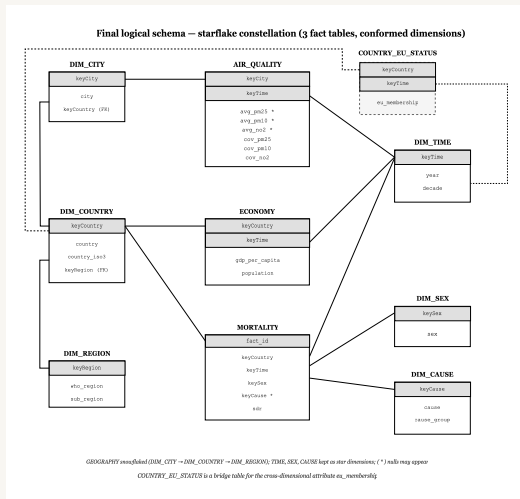
Fact	GEOGRAPHY	TIME	SEX	CAUSE
AIR_QUALITY	City	Year	-	-
ECONOMY	Country	Year	-	-
MORTALITY	Country	Year	Sex	Cause

■ Grain per fact and conformed dimensions

- Different grain per fact: AIR_QUALITY at **City** level (urban measurements), ECONOMY/MORTALITY at **Country** level (national data)
- Shared conformed dimensions: GEOGRAPHY and TIME (year → decade) enable cross-domain analysis

Implementation Details

- **Constellation structure** for the three fact tables.
- **Snowflake for Geography:**
The geographic hierarchy is normalized into three separate tables linked by foreign keys.
- **Star for the other dimensions:**
Time, Sex, and Cause are kept "flat" and denormalized in single tables.
- **Surrogate** keys assigned to each dimension table.



Why Snowflake for Geography

city → country → sub_region: avoids repeating the country name/ISO3 for hundreds of cities (redundancy), and handles the two different grain levels (city for air quality, country for economy/mortality) — a single flat DIM_GEOGRAPHY could not do this cleanly.

Why Star for Time, Sex, Cause

These are **small, static** tables (few rows, almost never updated): normalizing them like geography would add extra JOINS with no real benefit, since there is no significant redundancy to remove. Denormalization here is a pure win for reads.

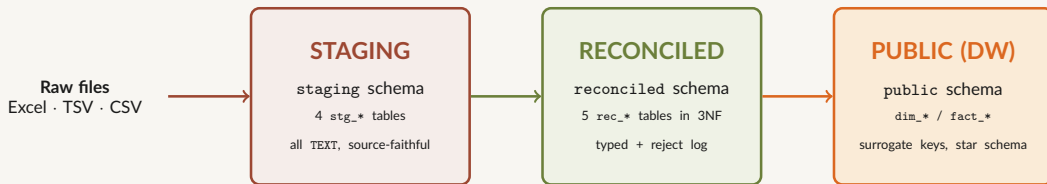
■ The problem

Only **3 specific diseases** belong to DIM_CAUSE. The total entry "*All causes*" is not a fourth cause: it is the overall total.

■ Solution: optional key

- "*All causes*" → `key_cause = NULL` in the MORTALITY fact
- Integrity guaranteed in PostgreSQL by **two partial unique indexes**:
`uq_mortality_cause` and `uq_mortality_total`
- They prevent double counting between totals and per-cause values

3-TIER ELT ARCHITECTURE



We adopt a **3-tier ELT**: raw data is loaded immediately, as-is, into the staging schema; cleaning and integration happen *afterwards*, with declarative SQL queries inside PostgreSQL, moving through the reconciled and public tiers.

Python's role changes: only *ingestion*, never *cleaning*

Python reads Excel/TSV/CSV files and writes them *exactly as they are* into the staging schema: no parsing, no casting. Everything stays TEXT on purpose — a NUMERIC cast would immediately fail on Eurostat flags ("31200 p", ": "); casting becomes a decision postponed to the next step.

Domain	Staging table	Source
Air quality	stg_who_aap	1 WHO Excel file
GDP	stg_eurostat_gdp	1 Eurostat TSV file
Population	stg_eurostat_pop	1 Eurostat TSV file
Mortality	stg_hfamdb_mortality	4 HFAMDB CSV files (UNION)

■ From *staging* to *reconciled*: cleaning and integration

- **Cleaning:** non-interpretable values are discarded; *wide* format (one year per column) → *long* format (one row per year)
- **Integration:** GDP and Population (2 separate Eurostat files) are merged into a single row per country-year, without losing data when one source does not cover a given country/year
- Tables in **3NF**: **natural** keys (e.g. `country_iso3`), constraints (PK, FK, CHECK), and specific types instead of the TEXT used in staging

reconciled.* table	Grain (1 row per ...)
rec_country	Country
rec_city	(Country, City)
rec_air_quality	(Country, City, Year)
rec_economy	(Country, Year)
rec_mortality	(Country, Year, Sex, Cause)

Every discarded row leaves a trace

Every rejection creates an explicit row in `reconciled.rec_reject_log`: table, key, reason, and timestamp — so we can reconstruct what was excluded and why, without re-running the pipeline.

Reason for rejection	Example
Eurostat aggregate, not a country	EU27_2020, EA20
ISO3 code outside geographic Europe	AFG, USA

`v_reject_summary`: the aggregated log

A view that groups the log by table and reason, counting occurrences: one query answers "how many rows were excluded, and why".

■ From reconciled to public: surrogate key lookup

The `reconciled.*` tables feed the dimensions and facts, replacing natural keys (ISO3, city, year) with surrogate keys. At the end of the ELT process, the `public` schema contains the fully populated fact constellation, ready for analytical queries.

From reconciled.*	To public.*
<code>rec_country</code>	<code>dim_region, dim_country</code>
<code>rec_city</code>	<code>dim_city</code>
<code>rec_time</code>	<code>dim_time</code>
<code>rec_air_quality</code>	<code>air_quality (fact)</code>
<code>rec_economy</code>	<code>economy (fact), country_eu_status</code>
<code>rec_mortality</code>	<code>mortality (fact)</code>

For mortality, the *"All causes"* row does not match any specific cause: `key_cause` automatically stays NULL.

■ Inconsistent country codes

WHO/HFAMDB use **ISO3** (3 letters), Eurostat uses **geo** with **2 letters** (2 exceptions: EL = Greece, UK = United Kingdom): a dedicated mapping table is needed to bring them all back to one common key.

- `sub_region` (Northern/Southern/Eastern/Western Europe): does not exist in the original datasets, defined by hand to enable geographic roll-up
- `cause_group` (e.g. *cardiopulmonary*, *cancer*): the 3 specific causes of death were grouped from scratch for a more natural drill-down
- Cleaning numbers: Eurostat flags ("31200 p" = provisional, ":" = missing) are cleaned before casting to numeric

ROLL-UP

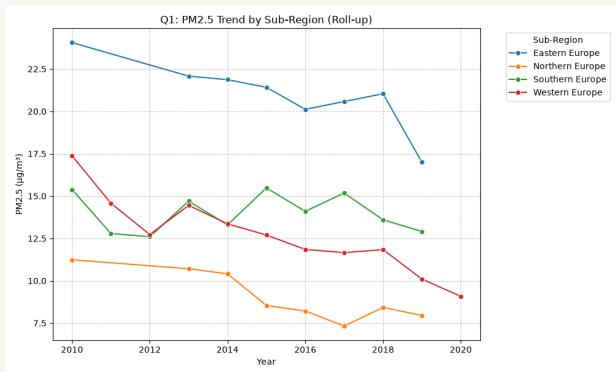
```
SELECT r.sub_region,
       t.year,
       COUNT(DISTINCT ci.key_city)      AS n_cities,
       ROUND(AVG(aq.avg_pm25), 2)      AS pm25,
       ROUND(AVG(aq.avg_pm10), 2)      AS pm10,
       ROUND(AVG(aq.avg_no2), 2)      AS no2
FROM   air_quality aq
JOIN   dim_city    ci ON ci.key_city    = aq.key_city
JOIN   dim_country co ON co.key_country = ci.key_country
JOIN   dim_region  r  ON r.key_region   = co.key_region
JOIN   dim_time    t  ON t.key_time     = aq.key_time
GROUP BY r.sub_region, t.year
ORDER BY r.sub_region, t.year;
```

Roll-up query that aggregates air quality data. Air quality is stored at city grain. We climb the geographic hierarchy city → country → sub_region and aggregate to the region level, by year.

ROLL-UP INSIGHT: PM2.5 TREND

Roll-up

Trend of PM2.5 levels over time across different sub-regions in Europe.



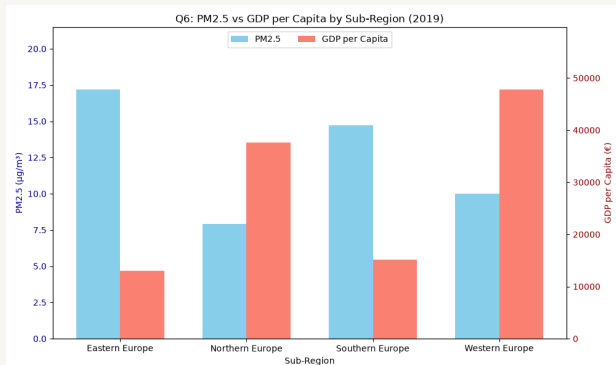
DRILL-ACROSS

```
WITH aq_country AS (
    SELECT ci.key_country, aq.key_time,
           AVG(aq.avg_pm25) AS pm25, AVG(aq.avg_no2) AS no2, COUNT(*) AS n_cities
    FROM   air_quality aq
    JOIN   dim_city ci ON ci.key_city = aq.key_city
    GROUP BY ci.key_country, aq.key_time
),
resp AS (
    SELECT m.key_country, m.key_time, m.sdr
    FROM   mortality m
    JOIN   dim_cause ca ON ca.key_cause = m.key_cause
    JOIN   dim_sex s ON s.key_sex = m.key_sex
    WHERE  ca.cause = 'Respiratory diseases' AND s.sex = 'ALL'
)
SELECT r.sub_region, t.year,
       COUNT(DISTINCT a.key_country) AS n_countries,
       SUM(a.n_cities) AS n_cities,
       ROUND(AVG(a.pm25)::numeric, 2) AS pm25,
       ROUND(AVG(re.sdr)::numeric, 2) AS respiratory_sdr,
       ROUND(AVG(e.gdp_per_capita)::numeric, 0) AS gdp_per_capita
FROM   aq_country a
JOIN   dim_country co ON co.key_country = a.key_country
JOIN   dim_region r ON r.key_region = co.key_region
JOIN   dim_time t ON t.key_time = a.key_time
LEFT JOIN resp re ON re.key_country = a.key_country AND re.key_time = a.key_time
LEFT JOIN economy e ON e.key_country = a.key_country AND e.key_time = a.key_time
GROUP BY r.sub_region, t.year
HAVING COUNT(DISTINCT a.key_country) >= 3
ORDER BY r.sub_region, t.year;
```

POLLUTION AND WEALTH: THE CONSTELLATION AT WORK

Drill-Across

Joining AIR_QUALITY and ECONOMY on the conformed dimensions GEOGRAPHY and TIME: in 2019 Eastern Europe has the highest PM2.5 ($17.2 \mu\text{g}/\text{m}^3$) and the lowest GDP per capita (€12,970) – Western Europe shows the exact opposite ($10.0 \mu\text{g}/\text{m}^3$, €47,767).



■ In summary

- An effective **ROLAP Constellation** architecture for connecting heterogeneous domains (air, economy, health)
- DFM design → Starflake logical schema → 3-tier ELT pipeline (staging → reconciled → public), reproducible and auditable
- Advanced OLAP analyses reveal the link between pollution, GDP, and mortality

Moving to the live demo